Xiayin Lou

Doctoral Researcher, Chair of Cartography and Visual Analytics, Technical University of Munich

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Education

Technical University of Munich Munich, Germany
Ph.D. in Cartography & Geoinformatics (Chair of Cartography and Visual Analytics) 2024–present

Peking University Beijing, China
M.Sc. in Cartography and Geographic Information Systems 2020–2023

Zhejiang A&F University Hangzhou, China
B.Eng. in Geographic Information Science 2016–2020

Academic Appointment

University of Salzburg Salzburg, Austria
Visiting Scholar, Department of Geoinformatics (hosted by Johannes Scholz) Oct.–Dec. 2026

Technical University of Munich Munich, Germany
Doctoral Researcher, Chair of Cartography and Visual Analytics 2024–present

Research Interests

GeoAI; trustworthy and uncertainty-aware geospatial AI; uncertainty quantification for spatial prediction; explainable spatial modeling; geographic bias; LLM-driven geospatial model discovery.

Publications

* Equal contribution † Corresponding author ⋆ Journal most-read list **Bold** indicates the author of this CV.

Refereed Journal Articles

1. **Lou, X.**, Luo, P.†, Li, Z., Gao, S., & Meng, L. (2025). GeoXCP: uncertainty quantification of spatial explanations in explainable AI. *International Journal of Geographical Information Science*. doi:10.1080/13658816.2025.2574900 (⋆ Top-5 most read)
2. **Lou, X.**, Luo, P.†, & Meng, L. (2025). GeoConformal prediction: a model-agnostic framework for measuring the uncertainty of spatial prediction. *Annals of the American Association of Geographers*, 115(8), 1971–1998. doi:10.1080/24694452.2025.2516091 (⋆ Top-10 most read)
3. **Lou, X.**, Sun, M.†, & Yang, S. (2022). A fine-grained navigation network construction method for urban environments. *International Journal of Applied Earth Observation and Geoinformation*, 113, 102994. doi:10.1016/j.jag.2022.102994
4. **Lou, X.**, Sun, M.†, Yang, H., & Yang, S. (2023). FINNCH: cooperative pursuit navigation for a pursuer team to capture a single evader in urban environments. *ISPRS International Journal of Geo-Information*, 12(12), 475. doi:10.3390/ijgi12120475

5. Yang, S., Sun, M., **Lou, X.**, Yang, H., & Liu, D. (2024). Nighttime thermal infrared image translation integrating visible images. *Remote Sensing*, 16(4), 666. doi:10.3390/rs16040666
6. Yang, S., Sun, M., **Lou, X.**, Yang, H., & Zhou, H. (2023). An unpaired thermal infrared image translation method using GMA-CycleGAN. *Remote Sensing*, 15(3), 663. doi:10.3390/rs15030663

Peer-Reviewed Conference Proceedings

1. **Lou, X.**, & Luo, P.† (2025). Towards the uncertainty-aware geospatial artificial intelligence. In *Proceedings of the 8th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery (GeoAI '25)*, 76–80.
2. **Lou, X.**, Luo, P.†, & Meng, L. (2025). Uncertainty-aware geospatial visual analytics. *Abstracts of the ICA*, 10. (32nd International Cartographic Conference, ICC 2025.)

Manuscripts Under Review & Preprints

1. **Lou, X.**, & Luo, P.† (2026). Weighted Bayesian conformal prediction. arXiv:2604.06464
2. Luo, P.*, **Lou, X.***, Zheng, Y., Zheng, Z., & Ermon, S. (2025). GeoEvolve: automating geospatial model discovery via multi-agent large language models. Under review, *International Conference on Learning Representations (ICLR) 2026*. arXiv:2509.21593

Awards, Grants & Scholarships

Grants & Travel Awards

- Conference Travel Grant, GeoAI 2026, *Ghent, Belgium* 2026
- Conference Travel Grant, International Cartographic Conference (ICC), *400 EUR* 2025

Scholarships & Honors

- May Fourth Scholarship, *Peking University* 2022
- Second-class Academic Scholarship, *Zhejiang A&F University* 2017–2019
- Zhejiang Province Government Scholarship, *Zhejiang Provincial Government* 2017

Talks & Presentations

- GeoXCP: uncertainty quantification of spatial explanations in XAI (oral), *GeoAI 2026, Ghent, Belgium* 2026
- Spatiotemporal Innovation Symposium 2025
- Next Generation Cartographer 2025

Teaching

- Spatial Visual Analytics — lecture on Trustworthy GeoAI, *M.Sc., Technical University of Munich* 2026

Student Mentoring

- Weiyi Gao (M.Sc. thesis), *Geospatial algorithm discovery driven by LLMs: spatial prediction model as an example* 2026–present

Open-Source Software

- GeoEvolve 2025
  LLM-based multi-agent framework for automatic geospatial model discovery.
- GeoXCP 2025
  Model-agnostic uncertainty quantification for spatial explainable AI.
- GeoCP (Python package `geoconformal`) 2025
  Model-agnostic uncertainty quantification for any spatial prediction model.

Media Coverage

- 36Kr, *Feature on* GeoEvolve — *multi-agent LLMs for geospatial model discovery* 2025

Professional Service

Journal Peer Review

- Annals of the American Association of Geographers
- International Journal of Geographical Information Science
- Computational Urban Science
- GIScience & Remote Sensing

Last updated: May 24, 2026